

### 3 Probability

#### What students need to learn:

Elementary probability.

Sample space. Exclusive and complementary events. Conditional probability.

Independence of two events.

Sum and product laws.

Understanding and use of

$$P(A') = 1 - P(A),$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

$$P(A \cap B) = P(A) P(B | A).$$

$$P(B | A) = P(B), P(A | B) = P(A),$$

$$P(A \cap B) = P(A) P(B).$$

Use of tree diagrams and Venn diagrams. Sampling with and without replacement.

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### 4 Correlation and regression

#### What students need to learn:

Scatter diagrams. Linear regression.

Explanatory (independent) and response (dependent) variables. Applications and interpretations.

The product moment correlation coefficient, its use, interpretation and limitations.

Calculation of the equation of a linear regression line using the method of least squares. Students may be required to draw this regression line on a scatter diagram.

Use to make predictions within the range of values of the explanatory variable and the dangers of extrapolation. Derivations will not be required. Variables other than  $x$  and  $y$  may be used. Linear change of variable may be required.

Derivations and tests of significance will not be required.

## 5 Discrete random variables

### What students need to learn:

The concept of a discrete random variable.

The probability function and the cumulative distribution function for a discrete random variable.

Simple uses of the probability function  $p(x)$  where  $p(x) = P(X = x)$ .  
Use of the cumulative distribution function:

$$F(x_0) = P(X \leq x_0) = \sum_{x \leq x_0} p(x)$$

Mean and variance of a discrete random variable.

Use of  $E(X)$ ,  $E(X^2)$  for calculating the variance of  $X$ .

Knowledge and use of

$$E(aX + b) = aE(X) + b,$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

The discrete uniform distribution.

The mean and variance of this distribution.

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## 6 The Normal distribution

### What students need to learn:

The Normal distribution including the mean, variance and use of tables of the cumulative distribution function.

Knowledge of the shape and the symmetry of the distribution is required. Knowledge of the probability density function is not required. Derivation of the mean, variance and cumulative distribution function is not required. Interpolation is not necessary. Questions may involve the solution of simultaneous equations.

## Unit S2

## Statistics 2

GCE Mathematics, GCE AS and GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional) A2 optional unit

### S2.1 Unit description

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The Binomial and Poisson distributions; continuous random variables; continuous distributions; samples; hypothesis tests.

### S2.2 Assessment information

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#### Prerequisites

A knowledge of the specification for S1 and its prerequisites and associated formulae, together with a knowledge of differentiation and integration of polynomials, binomial coefficients in connection with the binomial distribution and the evaluation of the exponential function is assumed and may be tested.

#### Examination

The examination will consist of one 1½ hour paper. The paper will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

#### Calculators

Students are expected to have available a calculator with at least the following keys: +, −, ×, ÷,  $\pi$ ,  $x^2$ ,  $\sqrt{x}$ ,  $\frac{1}{x}$ ,  $x^y$ ,  $\ln x$ ,  $e^x$ ,  $x!$ , sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

## Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

For the continuous random variable  $X$  having probability density function  $f(x)$ ,

$$P(a < X \leq b) = \int_a^b f(x) \, dx.$$

$$f(x) = \frac{dF(x)}{dx}.$$

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## 1 The Binomial and Poisson distributions

### What students need to learn:

The binomial and Poisson distributions.

Students will be expected to use these distributions to model a real-world situation and to comment critically on their appropriateness. Cumulative probabilities by calculation or by reference to tables.

Students will be expected to use the additive property of the Poisson distribution — eg if the number of events per minute  $\sim \text{Po}(\lambda)$  then the number of events per 5 minutes  $\sim \text{Po}(5\lambda)$ .

The mean and variance of the binomial and Poisson distributions.

No derivations will be required.

The use of the Poisson distribution as an approximation to the binomial distribution.

## 2 Continuous random variables

### What students need to learn:

The concept of a continuous random variable.

The probability density function and the cumulative distribution function for a continuous random variable.

Use of the probability density function  $f(x)$ , where

$$P(a < X \leq b) = \int_a^b f(x) \, dx.$$

Use of the cumulative distribution function

$$F(x_0) = P(X \leq x_0) = \int_{-\infty}^{x_0} f(x) \, dx.$$

The formulae used in defining  $f(x)$  will be restricted to simple polynomials which may be expressed piecewise.

Relationship between density and distribution functions.

$$f(x) = \frac{dF(x)}{dx}$$

Mean and variance of continuous random variables.

Mode, median and quartiles of continuous random variables.

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## 3 Continuous distributions

### What students need to learn:

The continuous uniform (rectangular) distribution.

Including the derivation of the mean, variance and cumulative distribution function.

Use of the Normal distribution as an approximation to the binomial distribution and the Poisson distribution, with the application of the continuity